VITYARTHI PROJECT

TOPIC: STUDENT STUDY PLANNER
NAME: GOKUL PRASAD. K
REG NO.: 25BAI10148
COURSE TITLE: FUNDAMENTALS IN AI AND ML
COURSE CODE: CSA2001
COURSE TYPE: FLIPPED COURSE
COURSE CREDITS: 4
FACULTY: Dr. SUBASH CHANDRA BOSE. J
SLOT: A21 +A22 +A23
SUBMISSION DATE: 30/03/2026

INTRODUCTION

PROJECT OVERVIEW:

In the current competitive academic environment, it has been a challenging task for students to manage their time effectively while studying different subjects and upcoming examinations. The purpose of this project, "Student Study Planner," is to design a personal and adaptive plan for students to manage their time effectively while consistent with their studies. The system has taken into consideration different critical factors such as difficulty level of the subject, exam dates, weaker aspects of the students, time of the day for studying, speed of studying, and day-to-day distractions for developing a day - wise plan for the students. Using the logic of artificial intelligence, the system has tried to make wise decisions to optimize the time for studying and effective time management for the students.

It also tracks performance, has a streak of completed tasks, and adjusts future schedules based on user feedback. Special features such as revision days, break plans, and motivational messages are also incorporated into the planner to make it more realistic and practical for the users. This project not only helps the users to effectively manage their time but also helps them develop self-discipline and improve their learning. By providing a more organized and flexible way of learning, the Student Study Planner helps the users to effectively achieve their academic goals without the stress of studying for exams.

PROBLEM STATEMENT:

It is difficult for the students to effectively plan their study schedules while preparing for different subjects, which vary in difficulty levels and exams. The current system of planning lacks the flexibility and efficiency required, along with the proper monitoring of the performance and poor prioritization of the subjects. The aim of this project is to address the problem by developing an interactive system for the students, which will be referred to as the "Student Study Planner" where the students can enter the details of the subjects, the dates for the exams, the difficulty level, weak subjects based on their performance, the hours for studying, the time of the day for studying, and the level of their performance. The system will generate a flexible day-wise plan for the students, considering the weak subjects based on the performance level, distractions, addition of revision days, breaks, motivational quotes, and time.

PROJECT OBJECTIVES:

- To allow students to input all relevant data, including subjects, difficulty levels, exam dates, weak subjects, study hours, and personal preferences.
- To generate an adaptive day - wise study schedule based on subject priority, exam urgency, weak areas, and learning speed.
- To track daily performance and maintain streaks, adjust future if tasks are missed.
- To highlight weak subjects and urgent topics, ensuring students focus on areas that need improvement.
- To provide practical features such as revision days, break planning, motivational messages, and time – slot suggestions.
- To help students optimize study hours, improve consistency, and reduce exam stress

FUNCTIONAL REQUIREMENTS:

- The system shall allow the user to enter subjects with difficulty levels and days left for exams.
- The system shall allow the user to input total study hours per day and number of planning days.
- The system shall allow the user to specify weak subjects, energy preference, and learning speed.
- The system shall allow the user to record daily performance (completed or not) and distraction levels.
- The system shall calculate subject priorities based on difficulty, urgency, and weak subjects.
- The system shall generate a day-wise adaptive study schedule with allocated study hours.
- The system shall adjust study hours dynamically based on performance and distractions.
- The system shall highlight weak subjects and upcoming exams for focused study.
- The system shall suggest time slots and revision days according to energy and urgency.
- The system shall display motivational messages and update plans for the next day automatically.

NON-FUNCTIONAL REQUIREMENTS:

- The system shall have a user - friendly interface for easy input of subjects, study hours, and daily performance.
- The system shall be platform -independent and run on any Python environment.
- The system shall process and generate the study plan quickly, with minimal delay.
- The system shall ensure data consistency, updating priorities, and schedules accurately each day.
- The system shall provide clear and readable output for each day's plan.
- The system shall be reliable, correctly adjusting schedules based on performance and distractions.
- The system shall maintain accuracy in calculating study hours, priorities, and streaks.
- The system shall be scalable, supporting any number of subjects and days.
- The system shall allow easy modification of input parameters without changing the code.
- The system shall be maintainable, with a clear structure to add new features like notifications or reports in the future.

PROGRAM FLOW:

- **Start the Program**
  The program begins by displaying the Student Study Planner system and prompting the user for input.
- **User Inputs**
  - Number of subjects and details for each subject (name, difficulty, days left for exams)
  - Total study hours per day and number of days to plan
  - Weak subject, energy preference (morning/evening), and learning speed (slow/fast)
  - Daily performance (completed or not) and distraction level for each day

 Priority Calculation

For each day, the system calculates a **priority score** for every subject based on:

- Subject difficulty
- Days left for exams (urgency)
- Weak subject status

 Study Hour Allocation

- Allocate study hours proportionally according to priority scores
- Adjust hours based on learning speed and weak subjects.
- Suggest time slots (morning/evening/flexible) based on energy preference and subject difficulty.

- **Display Day-wise Plan**
  - Show allocated study hours and time slots for each subject
  - Provide explanations for subject allocation
  - Display special notes: revision days, break planning, motivational messages
- **Performance Tracking and Updates**
  - Update streaks based on daily completion
  - Adjust study hours and priorities based on distraction levels and performance
  - Mark completed subjects and update days left for exams
- **Next Day Planning**
  - Automatically update the plan for the next day based on current performance and priorities
- **Repeat for All Days**
  - Continue the same process for the total number of days planned
- **Final Advice**
  - After completing all days, display tips and recommendations to help maintain consistency, focus on weak subjects, and optimize study time.
- **End Program**

RESULT:

```
Enter number of subjects:  3
Enter name of subject 1:  mathematics
Enter difficulty (easy/medium/hard):  hard
Enter days left for exam:  20
Enter name of subject 2:  physics
Enter difficulty (easy/medium/hard):  medium
Enter days left for exam:  18
Enter name of subject 3:  fundamentals in AI and ML
Enter difficulty (easy/medium/hard):  easy
Enter days left for exam:  15
Enter total study hours per day:  5
Enter number of days to plan:  5
When are you most active? (morning/evening):  morning
Learning speed (fast/slow):  fast
Enter weak subject (or 'none'):  none
```

```
enter performance and distraction for each day:
Day 1 - Did you complete plan? (yes/no):  yes
Day 1 - Distraction level (1-10):  3
Day 2 - Did you complete plan? (yes/no):  no
Day 2 - Distraction level (1-10):  8
Day 3 - Did you complete plan? (yes/no):  yes
Day 3 - Distraction level (1-10):  3
Day 4 - Did you complete plan? (yes/no):  no
Day 4 - Distraction level (1-10):  7
Day 5 - Did you complete plan? (yes/no):  yes
Day 5 - Distraction level (1-10):  2
```

```
STUDENT STUDY PLANNER SYSTEM
Total Subjects: 3
Planning Days: 5
Weak Subject: none
Energy Preference: morning
Learning Speed: fast

********************
DAY 1 STUDY PLAN
********************

--- Subject Allocation ---

Subject: mathematics
Study Hours: 2.19
Time Slot: Morning (High Energy)
Explanation:
- Hard subject
```

```
Subject: physics
Study Hours: 1.37
Time Slot: Flexible Time
Explanation:
- Medium subject

Subject: fundamentals in AI and ML
Study Hours: 1
Time Slot: Flexible Time
Explanation:
- Easy subject

--- Special Notes ---
Break Plan: 10 min break after 2 hours
```

```
--- Performance Tracking ---
Task Completed: YES
Current Study Streak: 1
Motivation: Keep it up!
Distraction Level: 3
--- Updating System for Next Day ---

********************
DAY 2 STUDY PLAN
********************

--- Subject Allocation ---

Subject: mathematics
Study Hours: 2.18
Time Slot: Morning (High Energy)
Explanation:
- Hard subject
```


```
Subject: physics
Study Hours: 1.37
Time Slot: Flexible Time
Explanation:
- Medium subject

Subject: fundamentals in AI and ML
Study Hours: 1
Time Slot: Flexible Time
Explanation:
- Easy subject

--- Special Notes ---
Break Plan: 10 min break after 2 hours
```

```
--- Performance Tracking ---
Task Completed: NO
Streak Reset
Motivation: Try to focus more tomorrow!
Distraction Level: 8
--- Updating System for Next Day ---

********************
DAY 3 STUDY PLAN
********************

--- Subject Allocation ---

Subject: mathematics
Study Hours: 2.18
Time Slot: Morning (High Energy)
Explanation:
- Hard subject
```

```
Subject: physics
Study Hours: 1.37
Time Slot: Flexible Time
Explanation:
- Medium subject

Subject: fundamentals in AI and ML
Study Hours: 1
Time Slot: Flexible Time
Explanation:
- Easy subject

--- Special Notes ---
Revision Day Activated
Break Plan: 10 min break after 2 hours
```

```
--- Performance Tracking ---
Task Completed: YES
Current Study Streak: 1
Motivation: Keep it up!
Distraction Level: 3
--- Updating System for Next Day ---

********************
DAY 4 STUDY PLAN
********************

--- Subject Allocation ---

Subject: mathematics
Study Hours: 2.18
Time Slot: Morning (High Energy)
Explanation:
- Hard subject
```

```
Subject: physics
Study Hours: 1.37
Time Slot: Flexible Time
Explanation:
- Medium subject

Subject: fundamentals in AI and ML
Study Hours: 1
Time Slot: Flexible Time
Explanation:
- Easy subject

--- Special Notes ---
Break Plan: 10 min break after 2 hours
```

```
--- Performance Tracking ---
Task Completed: NO
Streak Reset
Motivation: Try to focus more tomorrow!
Distraction Level: 7
--- Updating System for Next Day ---

********************
DAY 5 STUDY PLAN
********************

--- Subject Allocation ---

Subject: mathematics
Study Hours: 2.18
Time Slot: Morning (High Energy)
Explanation:
- Hard subject
```

```
Subject: physics
Study Hours: 1.37
Time Slot: Flexible Time
Explanation:
- Medium subject

Subject: fundamentals in AI and ML
Study Hours: 1
Time Slot: Flexible Time
Explanation:
- Easy subject

--- Special Notes ---
Break Plan: 10 min break after 2 hours
```

```
--- Performance Tracking ---
Task Completed: YES
Current Study Streak: 1
Motivation: Keep it up!
Distraction Level: 2
--- Updating System for Next Day ---

********************
FINAL ADVICE
********************
Stay consistent in your studies.
Focus more on weak subjects.
Revise regularly before exams.
Maintain a good study-life balance.
```

CONCLUSION:

The Student Study Planner is a valuable tool for students to manage study schedules effectively, prioritize weak subjects, and adapt plans according to performance, distractions, and exam urgency. It helps improve learning outcomes, maintain consistency, and reduce exam stress. Th e project's scope extends to any student seeking structured guidance and can be enhanced with reminders, progress tracking, or mobile access. Challenges faced include dynamically adjusting study hours, handling multiple user inputs, and simulating realistic study

priorities. Despite these, the system promotes disciplined study habits, optimized study hours, and a balanced approach, making it highly relevant for academic success.

REFERENCES:

1. VITYARTHI learning
2. Learning documentation from google
3. Peer feedback from:
    (I). Dr. Subash Chandra Bose J
    (II). S G Sudharshan – 25BAI10706
    (III). Raghul S K – 25BAI10288
    (IV). Dilip S – 25BAI10590
    (V). Bavesh p – 25BAI10093